THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE CHAPTER TEST SET B

Chapter 1: **Exploration: Entering the World of Chemistry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. While studying how a ball falls, a student ignores air resistance. This is: 1

(a) a deliberate simplification to understand the basic effect of gravity
(b) cheating in an experiment
(c) possible only for heavy balls
(d) a mistake that makes the model useless

2. Meghnad Saha's approach to stars treated the matter in a star as: 1

(a) a collection of living cells
(b) a liquid core only
(c) a hot gas, focusing on temperature and pressure
(d) a solid ball

3. The main advantage of using standard SI units everywhere is that: 1

(a) instruments become cheaper
(b) results can be compared and conversion errors are avoided
(c) measurements become exact
(d) calculations become unnecessary

4. The atomic theory explains: 1

(a) what 1 kg is **(b)** how molecules are formed
(c) how to measure time **(d)** why clouds look dark

5. Which statement would make a better scientific question about tomorrow's rain? 1

(a) Are clouds beautiful?
(b) What was the humidity the last time it rained?
(c) Is rain always good?
(d) Do you like rain?

6. A person at rest breathes about 12–15 times a minute. In one day the number of breaths is roughly: 1

(a) 2 crore **(b)** 20 thousand **(c)** 2 lakh **(d)** 2000

7. Mathematics in science is best seen as: 1

(a) a language to express and test relationships clearly
(b) unrelated to experiments
(c) only a tool for finding answers
(d) a hurdle to be memorised

8. The magnifying glass on the page numbers of the textbook stands for: 1

(a) units of measurement **(b)** estimation
(c) careful observation **(d)** choosing direction

9. Which activity is NOT part of a scientific method? 1

(a) Refining measurements
(b) Testing ideas with experiments
(c) Repeating observations
(d) Rejecting an idea on the basis of opinion alone

10. ‘Science does not need to be perfect to be reliable’. Which feature supports this? 1

(a) Only experts may question
(b) Theories can never be wrong
(c) Facts never change
(d) Predictions that fail lead to better ideas

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. State one consequence of mixing up pounds and kilograms as described in the aircraft incident. 2

12. Distinguish between a law and a theory in science. 2

13. A friend says, ‘A theory is just a guess.’ Respond in two sentences. 2

14. What is a prediction in science and how is it different from a guess? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. Differentiate between a law, a theory and a principle with one example of each. 3

16. ‘No scientific theory is ever final.’ Explain why this is a strength of science. 3

17. Explain how Meghna could test Varsha's prediction of rain with measurable evidence. Give three questions. 3

OR

A message says ‘Do not eat food during an eclipse, it becomes harmful’. How would you check this claim scientifically? 3

18. Why can forecasts go wrong even with good models? Mention two factors the weather depends on. 3

OR

Estimate the volume of air a person breathes in a day. Write the steps. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Explain how to estimate the litres of air a person breathes in a day. 5
(ii) Write the numbers used at each step.
(iii) Use the balloon method to check your answer and compare.

OR

Case study: A viral message claims that food should not be eaten during an eclipse because it becomes harmful. 5
(i) What is an eclipse?
(ii) Which measurements could test the claim?
(iii) What would a scientist conclude if no change is found?
(iv) Why is the habit of questioning important?

20. (i) What is estimation? 5
(ii) Estimate how much rice a family of four needs in a month, stating your assumptions.
(iii) Show how you judge whether the answer is sensible.

OR

Case study: A student wants to analyse a pressure cooker using ideas from different branches of science. 5
(i) Which physics idea is involved?
(ii) Which chemical or biological idea?
(iii) How does mathematics help?
(iv) Why can the cooker not be studied by one branch alone?

END OF QUESTION PAPER Total: 40 Marks